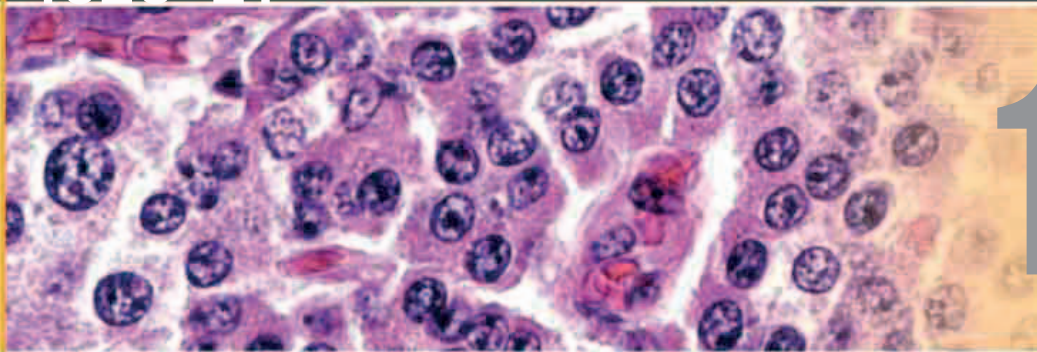




These clusters of cells, called Islets of Langerhans, are the source of the hormone insulin in the pancreas.

1 MILLION

THE NUMBER OF ISLETS OF LANGERHANS IN THE PANCREAS OF AN AVERAGE ADULT

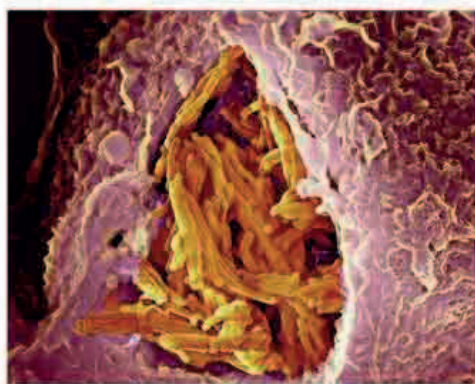
AMERICAN PHYSICIST ROBERT GODDARD (1882–1945)

had recently published a treatise promoting the idea of rocket space travel. On January 13, 1920, the *New York Times* ridiculed Goddard's views. A few years later he successfully launched a liquid-fueled rocket (see 1926).

In 1920, British physician Edward Mellanby (1884–1955) showed that a plain oatmeal diet caused a bone-softening deformity called **rickets**. He also found that the effects were eliminated by supplementing the diet with **cod-liver oil**. He deduced that cod liver must contain a factor that was needed for essential growth. In 1922, this factor was identified as **Vitamin D**.

BCG vaccine

Consisting of a harmless strain of bovine TB bacteria (orange), the BCG vaccine stimulates the body's immune system.



Later research revealed that exposure to the Sun's rays stimulates the formation of

Vitamin D in the body naturally.

In the same year, Arthur Eddington suggested that a **star's energy** comes from the conversion of hydrogen to helium by nuclear fusion. This was later supported by the findings of British astronomer Cecilia

Payne-Gaposchkin (1900–79), who showed that these elements predominated in a star.

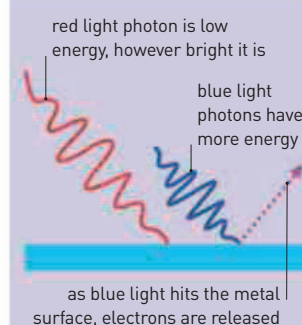
In Germany, botanist Hans Winkler (1877–1945) combined the terms "gene" (the particle of inheritance) and "chromosome" (solidified thread of genetic

material that becomes visible during cell division) to coin a new scientific term—**genome**. He used it in his book on biological reproduction to refer to a single set of genetic material.

In July 1921, the **BCG (Bacille Calmette–Guérin) vaccine** against tuberculosis (TB) was ready for its first human trial—15 years after Albert Calmette and Camille Guérin started developing it (see 1906). French physicians Benjamin Weill–Hallé (1875–1958) and Raymond Turpin (1895–1988) administered the vaccine orally to a newborn infant whose mother had died of TB. It was the start of a successful vaccination program that would extend worldwide.

In August, Canadian biologists Frederick Banting (1891–1941) and Charles Best (1899–1978) **isolated insulin**. It was known that the pancreas produced a secretion that regulated sugar levels in the body; the active ingredient had been called "insuline." However, the pancreas also produced digestive enzymes, and the destructive influence of these had always interfered with the extraction of insulin. Banting developed a way of tying off the digestive ducts so that insulin could be retrieved.

In 1921, the **Nobel Committee for Physics** decided that none of the nominations satisfied the prize



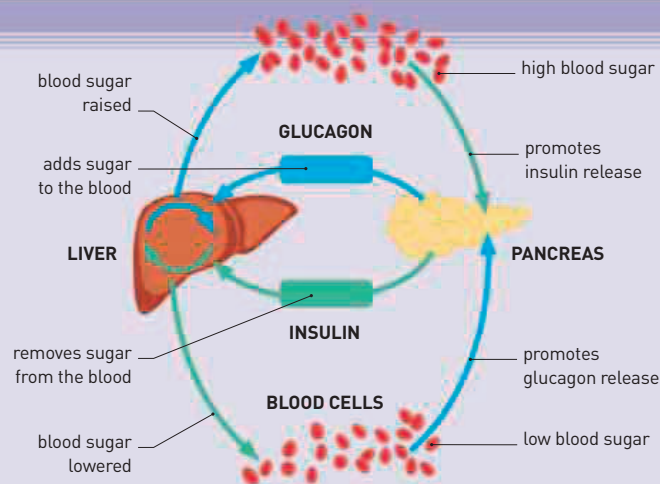
PHOTOELECTRIC EFFECT

The emission of electrons from a metallic surface when light shines on it is called the photoelectric effect. Albert Einstein explained it in terms of light particles with distinct quantities of energy called photons. He said that photons of red light (long wavelength) do not have enough energy to allow electrons to be emitted, but photons of blue light (short wavelength) do.

criteria, and carried it over to the next year. In 1922, **Albert Einstein** received the Nobel Prize for 1921, nearly two decades after his "miracle year," 1905, when he had published several groundbreaking theories. Einstein was awarded the prize for his services to Theoretical Physics and especially for his discovery of the law of the **photoelectric effect**.

BLOOD SUGAR REGULATION

The pancreas releases insulin when blood sugar concentration rises, such as after a meal. This prompts the liver to convert sugar to stored carbohydrate, causing the blood sugar to drop. When this happens, another hormone, glucagon, is released, prompting the liver to break down stored carbohydrate and release more sugar. The combined effect of the two hormones regulates the level of sugar in the blood.



January 13, 1920 The *New York Times* ridicules Robert Goddard's idea of a rocket

February 27, 1920 Edward Mellanby describes how rickets is caused by deficiency of a vitamin, later identified as Vitamin D

June 3, 1920 Ernest Rutherford coins the term **proton**

July 1920 US naturalists Wrightmann Garner and Harry Allard describe how the length of the day affects flowering

October 1920 British biochemist Jack Cecil Drummond devises a system of naming vitamins using letters of the alphabet

August 1921 Frederick Banting and Charles Best discover a hormone—**insulin**—that affects blood sugar levels

April 26, 1920 Smithsonian Museum of Natural History, Washington, DC, hosts the Great Debate between astronomers Harlow Shapley and Heber Curtis on the scale of the Universe

August 24, 1920 Arthur Eddington suggests that a star's energy comes from the nuclear fusion of hydrogen to form helium

1920 Hans Winkler coins the term **genome**

July 18, 1921 First BCG vaccine is administered as protection against tuberculosis

December 1921 German-American pharmacologist Otto Loewi describes a **chemical neurotransmitter** in the nervous system: acetylcholine